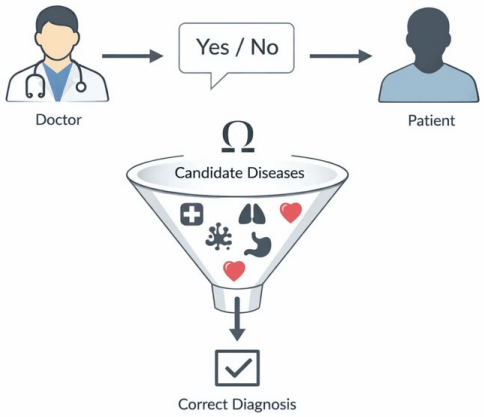
Replicating Tree Search and MCTS with Feedback for Interactive Diagnosis

INFO 7375_Assignment2_Yubo Wang

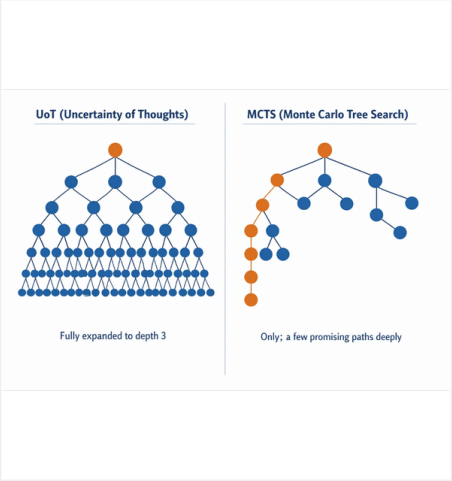
- Papers: UoT (Hu et al., 2024) + MISQ-HF (Chopra & Shah, 2025)
- Problem: Diagnose candidate diseases Ω in fewest yes/no turns
- Challenge: Choosing the optimal next question (Planning vs. MCTS)
- Project: Reproduce DP, UoT, MISQ, MISQ-HF on local small models



Doctor → Question → Patient → Yes/No → Narrow Ω

Methods: DP → UoT → MISQ → MISQ-HF

Method	Mechanism	Pros
DP	LLM asks freely	Simple, no overhead
UoT	Full tree (depth 3)	Future-aware
MISQ	Shared MCTS tree	Efficient, reuses work
MISQ-HF	MISQ + Feedback	Learns from success



UoT (Full) vs MISQ (Selective/Shared)

UCT Formula

$$UCT(v) = Q_{(v)} + C\sqrt{\frac{\ln(N_P)}{Nv}}$$

Feedback Bonus

$$bonus = \beta \times max(R, 0) \times \gamma$$

Setup & Implementation

Environment & Models

Dataset	DX (104 patients, $\|\Omega\|=5$)
Local Models	Llama 3.1:8b, Qwen 2.5:7b
Papers use	GPT-4, GPT-3.5, Llama 70B
Mode	Self-play (Doctor & Patient)

Parameters

Max turns / Depth	6 / 3 (UoT)
MCTS Iterations	10
Exploration C	0.2
Feedback β / γ / τ	0.2 / 0.9 / 0.9

Evaluation Metrics

SR
Success Rate (%)

MSC
Mean Turns (Success)

QGC
LLM Question Calls

Results — Tree Search Helps, Feedback Helps Speed

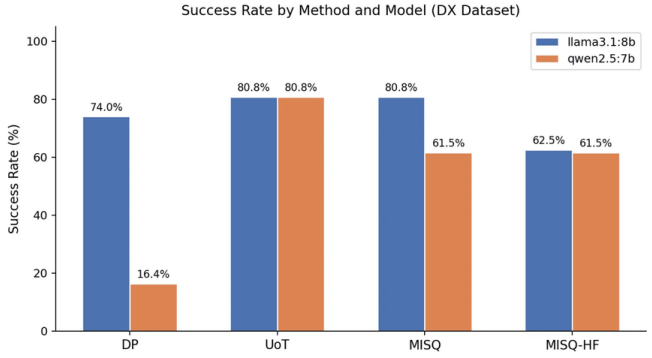
Main Results (Llama 3.1:8b)

Method	SR (%)	MSC (Turns)	QGC (Calls)
DP	74.04%	4.99	0
UoT	80.77%	3.00	8.00
MISQ	80.77%	3.00	0.03
MISQ-HF	62.50%	2.71	0.05

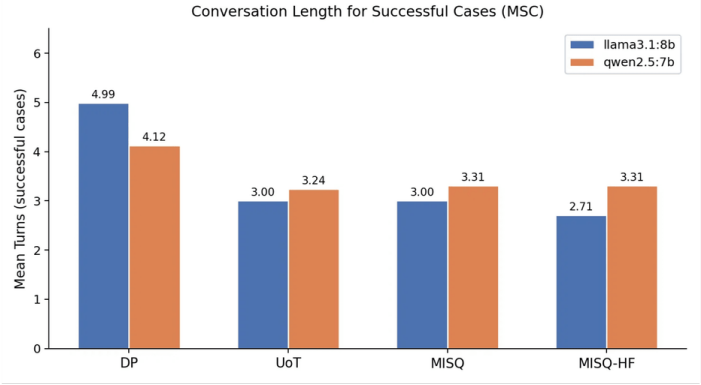
Paper Comparison (DX Dataset)

Method	Paper SR (GPT-4)	Our SR (8B)
DP	~75%	74.04%
UoT	82%	80.77%
MISQ-HF	87.5%	62.50%

* DP matches paper; UoT gap due to model size (8B vs GPT-4).



Success Rate (SR) by Method & Model



Mean Successful Turns (MSC)

Findings & Takeaways

Key Findings

- **Tree search helps a lot** — DP→UoT jumps Qwen from 16% to 81% SR.
- **MISQ is efficient** — Shared tree results in near-zero QGC overhead.
- **Feedback helps speed** — MISQ-HF achieves best MSC (2.71 turns).

Future Directions

- **Shared tree trade-off** — Saves compute but can amplify early errors.
- **Scalability** — Test on larger Ω where feedback signal accumulates better.
- **Next Steps** — Constrained-set Ωc and stronger base models.